

Q 1

1987

Find the equation of the tangent to the curve

$$y = 3x + e^{-x}$$

at the point where $x = 0$.

i) $y = 3x + e^{-x}$
 at $x = 0$, $y = 3(0) + e^{-0}$
 $= 0 + 1$
 $= 1$

$\therefore$ Point is $(0, 1)$

$$\frac{dy}{dx} = 3 - e^{-x}$$

at $x = 0$, $\frac{dy}{dx} = 3 - e^{-0}$
 $= 3 - 1$
 $= 2$

Equation of tangent at $(0, 1)$ with gradient = 2 is given by

$y - y_1 = m(x - x_1)$
 i.e. $y - 1 = 2(x - 0)$
 $y - 1 = 2x$
 $y = 2x + 1$.

Q 2

1988

Differentiate: xe^{2x}

$$\begin{aligned} \frac{d}{dx}(xe^{2x}) &= x \cdot \frac{d}{dx}(e^{2x}) + e^{2x} \cdot \frac{d}{dx}(x) \\ &= x \cdot 2e^{2x} + e^{2x} \cdot 1 \\ &= 2xe^{2x} + e^{2x} \\ &= e^{2x}(2x + 1). \end{aligned}$$

Q 3

1995

Differentiate: $4e^{2x}$

$$\begin{aligned} \frac{d}{dx}(4e^{2x}) &= 4e^{2x} \cdot \frac{d}{dx}(2x) \\ &= 4e^{2x} \cdot 2 \\ &= 8e^{2x}. \end{aligned}$$

Q 4

1996

Differentiate: xe^{2x}

$$\begin{aligned} \frac{d}{dx}(xe^{2x}) &= x \cdot \frac{d}{dx}(e^{2x}) + e^{2x} \cdot \frac{d}{dx}(x) \\ &= x \cdot 2e^{2x} + e^{2x} \cdot 1 \\ &= 2xe^{2x} + e^{2x}. \end{aligned}$$

Q 5

2001

Differentiate $\frac{x}{e^x}$.

$$\begin{aligned} \frac{d}{dx}\left(\frac{x}{e^x}\right) &= \frac{e^x \cdot 1 - xe^x}{(e^x)^2} \quad (\text{using quotient rule}) \\ &= \frac{e^x(1 - x)}{e^{2x}} \\ &= \frac{1 - x}{e^x}. \end{aligned}$$



Q 6
2002

 Find the equation of the tangent to $y = e^{2x}$ at the point $(0, 1)$.

$$y = e^{2x}$$

$$\frac{dy}{dx} = 2e^{2x}$$

$$\text{at } x = 0, \frac{dy}{dx} = 2e^0 = 2$$

 $\therefore$ Equation of tangent at $(0, 1)$ with gradient 2

$$\text{is } y - 1 = 2(x - 0)$$

$$y - 1 = 2x$$

$$y = 2x + 1.$$

Q 7
2002

 Differentiate $(2e^x - 4)^9$

$$\frac{d}{dx} [(2e^x - 4)^9]$$

$$= 9(2e^x - 4)^8 \cdot \frac{d}{dx} (2e^x - 4)$$

$$= 9(2e^x - 4)^8 \cdot (2e^x)$$

$$= 18e^x(2e^x - 4)^8.$$

Q 8
2005

 Find the coordinates of the point P on the curve $y = 2e^x + 3x$ at which the tangent to the curve is parallel to the line $y = 5x - 3$.

$$y = 2e^x + 3x$$

$$\frac{dy}{dx} = 2e^x + 3.$$

 Since the tangent is parallel to $y = 5x - 3$, then gradient of the tangent is 5.

$$\text{ie. } \frac{dy}{dx} = 5$$

$$2e^x + 3 = 5$$

$$2e^x = 2$$

$$e^x = 1$$

$$\therefore x = 0.$$

$$\text{When } x = 0, y = 2e^0 + 3(0) = 2.$$

 $\therefore$ The coordinates are $P(0, 2)$.

Q 9
2007

 Differentiate $\frac{2x}{e^x + 1}$

$$(2)(a)(i) \text{ let } y = \frac{2x}{e^x + 1}$$

$$\frac{dy}{dx} = \frac{2(e^x + 1) - 2xe^x}{(e^x + 1)^2}$$

$$= \frac{2e^x - 2xe^x + 2}{(e^x + 1)^2}$$

$$= \frac{2(e^x - xe^x + 1)}{(e^x + 1)^2}$$



extending calculus BOs

Part A: Differentiating expo



NAME: _____

Q10

Let $f(x) = 1 + e^x$.

2010

Show that $f(x) \times f(-x) = f(x) + f(-x)$.

Q4d $f(x) = 1 + e^x$

$$f(x) \times f(-x) = (1 + e^x)(1 + e^{-x}) = 1 + e^x + e^{-x} + 1 = f(x) + f(-x)$$

Q11

Differentiate $(3 + e^{2x})^5$

2012

$$\frac{d}{dx}(3 + e^{2x})^5 = 5(3 + e^{2x})^4(2e^{2x}) = 10e^{2x}(3 + e^{2x})^4$$

Q12

Differentiate $(e^x + x)^5$

2015

$$\frac{d}{dx}(e^x + x)^5 = 5(e^x + 1)(e^x + x)^4$$

Q13

The line $y = mx$ is a tangent to the curve $y = e^{2x}$ at a point P .

2014

(i) Sketch the line and the curve on one diagram.

1

(ii) Find the coordinates of P .

3

(iii) Find the value of m .

1

$$mx = e^{2x} \dots (1)$$

At P , the two functions have the same gradient: $m = 2e^{2x} \dots (2)$

$$\therefore 2e^{2x} \cdot x = e^{2x}$$

$$e^{2x}(2x - 1) = 0$$

$$\therefore x = \frac{1}{2}$$

$$\therefore P\left(\frac{1}{2}, e\right)$$

Q14

What is the derivative of e^{x^2} ?

2017

A. $x^2 e^{x^2 - 1}$

B. $2xe^{2x}$

C. $2xe^{x^2}$

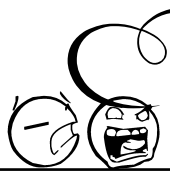
D. $2e^{x^2}$

Q15

2018

Differentiate $\frac{e^x}{x+1}$.

$$\begin{aligned}\frac{d}{dx}\left(\frac{e^x}{x+1}\right) &= \frac{(x+1) \times e^x - e^x \times 1}{(x+1)^2} \\ &= \frac{xe^x + e^x - e^x}{(x+1)^2} \\ &= \frac{xe^x}{(x+1)^2}\end{aligned}$$



extending calculus BOs

Part B: Differentiating LOG

2unit

NAME: _____

Q 1

1987

Differentiate: $\frac{\log_e x}{x}$.

$$\begin{aligned} \frac{d}{dx} \left(\frac{\log_e x}{x} \right) &= \frac{x \cdot \frac{d}{dx} (\log_e x) - \log_e x \cdot \frac{d}{dx} (x)}{x^2} \\ &= \frac{x \cdot \frac{1}{x} - \log_e x \cdot 1}{x^2} \\ &= \frac{1 - \log_e x}{x^2} \end{aligned}$$

Q 2

1989

- (i) Find the equation of the tangent to the curve $y = 3 \ln x + 2$ at the point where $x = 1$.
(ii) Find the slope of the normal to the curve at the point where $x = 1$.

(i) $y = 3 \ln x + 2$

$$\frac{dy}{dx} = 3 \cdot \frac{1}{x}$$

$$= \frac{3}{x}$$

$$= 3 \text{ at } x = 1$$

$$\text{at } x = 1, y = 3 \ln 1 + 2$$

$$= 3(0) + 2$$

$$= 2$$

Equation of tangent through (1, 2) with gradient = 3 is

$$y - 2 = 3(x - 1)$$

$$y - 2 = 3x - 3$$

$$y = 3x - 1.$$

- (ii) The tangent is perpendicular to the normal,

$$\therefore m_1 \times m_2 = -1$$

where m_1 and m_2 are the slopes of the tangent and normal respectively.

$$\therefore 3 \times m_2 = -1$$

$$m_2 = -\frac{1}{3}$$

$\therefore$ Slope of the normal is $-\frac{1}{3}$.

Q 3

1990

Differentiate: $x \ln x$.

$$\begin{aligned} \frac{d}{dx} (x \ln x) &= x \cdot \frac{d}{dx} (\ln x) + \ln x \cdot \frac{d}{dx} (x) \\ &= x \left(\frac{1}{x} \right) + \ln x (1) \\ &= 1 + \ln x. \end{aligned}$$

Q 4

1991

Differentiate: $\frac{\ln x}{x}$

$$\begin{aligned} \frac{d}{dx} \left(\frac{\ln x}{x} \right) &= \frac{x \cdot \frac{d}{dx} (\ln x) - \ln x \cdot \frac{d}{dx} (x)}{x^2} \\ &= \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2} \\ &= \frac{1 - \ln x}{x^2}. \end{aligned}$$

Q 5

1991

Differentiate: $\ln(5x - 1)$

$$\begin{aligned} \frac{d}{dx} [\ln(5x - 1)] &= \frac{1}{5x - 1} \cdot \frac{d}{dx} (5x - 1) \\ &= \frac{1}{5x - 1} \cdot 5 \\ &= \frac{5}{5x - 1}. \end{aligned}$$

Q 6
1992

 Differentiate: $x^2 \ln x$

$$\begin{aligned} \frac{d}{dx}(x^2 \ln x) &= x^2 \cdot \frac{d}{dx}(\ln x) + \ln x \cdot \frac{d}{dx}(x^2) \\ &= x^2 \cdot \frac{1}{x} + \ln x \cdot 2x \\ &= x + 2x \ln x \\ &= x(1 + 2 \ln x). \end{aligned}$$

Q 7
1995

 Differentiate: $x \log_e x$.

$$\begin{aligned} \frac{d}{dx}(x \log_e x) &= x \cdot \frac{d}{dx}(\log_e x) + \log_e x \cdot \frac{d}{dx}(x) \\ &= x \left(\frac{1}{x} \right) + \log_e x \cdot 1 \\ &= 1 + \log_e x. \end{aligned}$$

Q 7
2000

 Find the equation of the tangent to the curve $y = 2 \log_e x$ at $(1, 0)$.

$$y = 2 \log_e x, \quad \frac{dy}{dx} = \frac{2}{x}.$$

$$\text{At } x = 1, \quad \frac{dy}{dx} = 2.$$

$$\therefore \text{Equation of the tangent is } y - 0 = 2(x - 1) \\ y = 2x - 2.$$

Q 8
2001

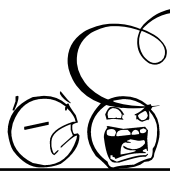
 Differentiate $\ln(x^2 - 9)$

$$\frac{d}{dx} [\ln(x^2 - 9)] = \frac{2x}{x^2 - 9}.$$

Q 9
2004

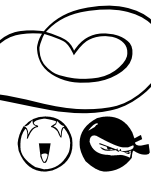
 Differentiate $x^2 \log_e x$

$$\begin{aligned} \frac{d}{dx}(x^2 \log_e x) &= x^2 \times \frac{1}{x} + \log_e x \times 2x \\ &= x + 2x \log_e x. \end{aligned}$$



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Part B: Differentiating LOG



NAME: _____

Q10

2005

Find the equation of the tangent to $y = \log_e x$ at the point $(e, 1)$

$$y = \log_e x \quad (e, 1)$$

$$\frac{dy}{dx} = \frac{1}{x} \quad \text{When } x = e, \quad \frac{dy}{dx} = \frac{1}{e}$$

Equation of tangent:

$$y - 1 = \frac{1}{e}(x - e)$$

$$ey - e = x - e \quad \left(\text{or } y = \frac{x}{e} \right)$$

Q11

2017

Differentiate $x^3 \ln x$.

$$\frac{d}{dx}(x^3 \times \ln x)$$

$$= x^3 \times \frac{1}{x} + 3x^2 \times \ln x$$

$$= x^2 + 3x^2 \times \ln x$$

$$u = x^3$$

$$\frac{du}{dx} = 3x^2$$

$$v = \ln x$$

$$\frac{dv}{dx} = \frac{1}{x}$$

Q 1

 Differentiate: $x \sin x$
2002

$$\begin{aligned}\frac{d}{dx}(x \sin x) &= x \cdot \frac{d}{dx}(\sin x) + \sin x \cdot \frac{d}{dx}(x) \\ &= x \cos x + \sin x \cdot 1 \\ &= x \cos x + \sin x.\end{aligned}$$

Q 2

 Differentiate $3x + \tan x$ with respect to x .

2003

$$\frac{d}{dx}(3x + \tan x) = 3 + \sec^2 x.$$

Q 3

 Differentiate with respect to x : $x^2 \sin x$.

2003

$$\begin{aligned}\frac{d}{dx}(x^2 \sin x) \\ &= x^2 \cdot \frac{d}{dx}(\sin x) + \sin x \cdot \frac{d}{dx}(x^2) \\ &= x^2 \cdot \cos x + \sin x \cdot 2x \\ &= x^2 \cos x + 2x \sin x.\end{aligned}$$

Q 4

 Differentiate with respect to x : $(1 + \sin x)^5$.

2002

$$\begin{aligned}\frac{d}{dx}(1 + \sin x)^5 &= 5(1 + \sin x)^4 \times (\cos x) \\ &= 5 \cos x (1 + \sin x)^4.\end{aligned}$$

Q 5

 Differentiate with respect to x : $x \tan x$
2006

$$\begin{aligned}y &= x \tan x \therefore u = x, u' = 1 \\ v &= \tan x, v' = \sec^2 x \\ \frac{dy}{dx} &= \tan x(1) + x \sec^2 x \\ &= \tan x + x \sec^2 x\end{aligned}$$

Q 6

 Differentiate with respect to x : $\frac{\sin x}{x+1}$
2006

$$\begin{aligned}y &= \frac{\sin x}{x+1} \therefore u = \sin x, u' = \cos x \\ v &= x+1, v' = 1 \\ \frac{dy}{dx} &= \frac{(x+1)\cos x - (\sin x)1}{(x+1)^2} \\ &= \frac{(x+1)\cos x - \sin x}{(x+1)^2}\end{aligned}$$



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Part C: Differentiating Trig

Unit

NAME: _____

Q 7

2006

Find the equation of the tangent to the curve $y = \cos 2x$ at the point whose x -coordinate is $\frac{\pi}{6}$.

$$y = \cos 2x$$

$$\text{at } x = \frac{\pi}{6} \therefore y = \cos \frac{2 \times \pi}{6} = \frac{1}{2}$$

$$\frac{dy}{dx} = -2 \sin 2x$$

$$\therefore \text{at } x = \frac{\pi}{6}, \frac{dy}{dx} = -2 \sin \left(2 \times \frac{\pi}{6} \right)$$

$$\frac{dy}{dx} = -2 \times \frac{\sqrt{3}}{2} = -\sqrt{3}$$

Using the point gradient formula

$$y - y_1 = m(x - x_1)$$

$$y - \frac{1}{2} = -\sqrt{3} \left(x - \frac{\pi}{6} \right)$$

$$2y - 1 = -2\sqrt{3} \left(x - \frac{\pi}{6} \right)$$

$$2y - 1 = -2\sqrt{3}x + \frac{\sqrt{3}\pi}{3}$$

$$6y - 3 = -6\sqrt{3}x + \sqrt{3}\pi$$

$$6\sqrt{3}x + 6y - 3 - \sqrt{3}\pi = 0$$

Q 8

2007

Differentiate with respect to x : $(1 + \tan x)^{10}$.

$$\text{let } y = (1 + \tan x)^{10}$$

$$\frac{dy}{dx} = 10(1 + \tan x)^9 \sec^2 x$$

$$= 10 \sec^2 x (1 + \tan x)^9$$

Q 9

2007

The point $P(\pi, 0)$ lies on the curve $y = x \sin x$. Find the equation of the tangent to the curve at P .

(c) $y = x \sin x$, using the product rule

$$\frac{dy}{dx} = \sin x + x \cos x$$

$$\text{at } x = \pi, \frac{dy}{dx} = \sin \pi + \pi \cos \pi = -\pi$$

$$y - 0 = -\pi(x - \pi)$$

$$y = -\pi x + \pi^2$$

$$x\pi + y - \pi^2 = 0$$

Q10

2008

Differentiate with respect to x : $\frac{\sin x}{x+4}$.

$$\frac{d}{dx} \left(\frac{\sin x}{x+4} \right) = \frac{(x+4)\cos x - \sin x}{(x+4)^2}$$

Q11

2009

Differentiate with respect to x : $x \sin x$

$$\frac{d}{dx} (x \sin x) = \sin x + x \cos x \text{ by the product rule.}$$

Q12

2010

Differentiate $x^2 \tan x$ with respect to x .

Q1e By the product rule:

$$\frac{d}{dx} (x^2 \tan x) = \left(\frac{d}{dx} x^2 \right) (\tan x) + x^2 \frac{d}{dx} \tan x = 2x \tan x + x^2 \sec^2 x$$

Q13

2010

Differentiate $\frac{\cos x}{x}$ with respect to x .

Q2a By the quotient rule:

$$\frac{d}{dx} \left(\frac{\cos x}{x} \right) = \frac{x(-\sin x) - \cos x}{x^2} = \frac{-x \sin x - \cos x}{x^2}$$



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Part C: Differentiating Trig



Unit

NAME: _____

Q14

2011

Differentiate $\frac{x}{\sin x}$ with respect to x .

$$\frac{d}{dx} \left(\frac{x}{\sin x} \right) = \frac{\sin x - x \cos x}{\sin^2 x}$$

Q15

2012

Differentiate with respect to x : $\frac{\cos x}{x^2}$

$$\begin{aligned} \text{Q12aii } \frac{d}{dx} \left(\frac{\cos x}{x^2} \right) &= \frac{\left(x^2 \right) \left(\frac{d}{dx} \cos x \right) - (\cos x) \left(\frac{d}{dx} x^2 \right)}{x^4} \\ &= \frac{x^2(-\sin x) - 2x \cos x}{x^4} = -\frac{x(x \sin x + 2 \cos x)}{x^4} = -\frac{x \sin x + 2 \cos x}{x^3} \end{aligned}$$

Q16

2013

What is the derivative of $\frac{x}{\cos x}$?

(A) $\frac{\cos x + x \sin x}{\cos^2 x}$

(B) $\frac{\cos x - x \sin x}{\cos^2 x}$

(C) $\frac{x \sin x - \cos x}{\cos^2 x}$

(D) $\frac{-x \sin x - \cos x}{\cos^2 x}$

$$\frac{d}{dx} \left(\frac{x}{\cos x} \right) = \frac{\cos x - x(-\sin x)}{\cos^2 x} = \frac{\cos x + x \sin x}{\cos^2 x}$$

A

Q17

2013

Differentiate $(\sin x - 1)^8$.

$$\frac{d}{dx} (\sin x - 1)^8 = 8 \cos x (\sin x - 1)^7$$

Q18

2016

What is the derivative of $\ln(\cos x)$?

(A) $-\sec x$

(B) $-\tan x$

(C) $\sec x$

(D) $\tan x$

Q19

2016

Find the gradient of the tangent to the curve $y = \tan x$ at the point where $x = \frac{\pi}{8}$.

Give your answer correct to 3 significant figures.

$$y = \tan x$$

$$\frac{dy}{dx} = \sec^2 x$$

$$\text{When } x = \frac{\pi}{8}, \frac{dy}{dx} = \sec^2 \left(\frac{\pi}{8} \right)$$

$$= \frac{1}{\cos^2 \frac{\pi}{8}}$$

$$\approx 1.17$$

$\therefore$ Gradient of tangent is 1.17.



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Part C: Differentiating Trig



Q20

2017

Differentiate $\frac{\sin x}{x}$.

$$\frac{d}{dx}\left(\frac{\sin x}{x}\right)$$

$$= \frac{x(\cos x) - \sin x(1)}{x^2}$$

$$= \frac{x \cos x - \sin x}{x^2}$$

$$u = \sin x$$

$$\frac{du}{dx} = \cos x$$

$$v = x$$

$$\frac{dv}{dx} = 1$$

Q21

2017

What is the derivative of $\sin(\ln x)$, where $x > 0$?

A. $\cos\left(\frac{1}{x}\right)$

B. $\cos(\ln x)$

C. $\cos\left(\frac{\ln x}{x}\right)$

D. $\frac{\cos(\ln x)}{x}$

Q22

2018

Differentiate $x^2 \tan x$.

$$\begin{aligned}\frac{d}{dx}(x^2 \tan x) &= x^2 \times \sec^2 x + \tan x \times (2x) \\ &= x(x \sec^2 x + 2 \tan x)\end{aligned}$$

Q23

2018

Find the equation of the tangent to the curve $y = \cos 2x$ at $x = \frac{\pi}{6}$.

$$y = \cos 2x$$

$$y' = -2 \sin 2x$$

$$M_T = -2 \sin\left(\frac{2\pi}{6}\right)$$

$$M_T = -2 \sin\left(\frac{\pi}{3}\right)$$

$$= -\sqrt{3} \text{ at } x = \frac{\pi}{6}$$

$$\text{when } x = \frac{\pi}{6}, y = \cos\left(\frac{2\pi}{6}\right)$$

$$= \frac{1}{2}$$

Equation of tangent

$$y - \frac{1}{2} = -\sqrt{3}\left(x - \frac{\pi}{6}\right)$$

$$y - \frac{1}{2} = -\sqrt{3}x + \frac{\sqrt{3}\pi}{6}$$

$$y = -\sqrt{3}x + \frac{\sqrt{3}\pi}{6} + \frac{1}{2}$$



Q 1

1987

Find the exact value of: $\int_0^1 e^{2x} dx$;

$$\begin{aligned}\int_0^1 e^{2x} dx &= \left[\frac{e^{2x}}{2} \right]_0^1 \\ &= \frac{1}{2}(e^2 - e^0) \\ &= \frac{1}{2}(e^2 - 1)\end{aligned}$$

Q 2

1989

Evaluate $\int_0^1 (e^{3x} + 1) dx$

$$\begin{aligned}\int_0^1 (e^{3x} + 1) dx &= \left[\frac{e^{3x}}{3} + x \right]_0^1 \\ &= \left(\frac{e^3}{3} + 1 \right) - \left(\frac{e^0}{3} + 0 \right) \\ &= \frac{e^3}{3} + 1 - \frac{1}{3} \\ &= \frac{e^3}{3} + \frac{2}{3} \\ &= \frac{1}{3}(e^3 + 2).\end{aligned}$$

Q 3

1990

Find: $\int_0^1 (1 + e^{-x}) dx$.

$$\begin{aligned}\int_0^1 (1 + e^{-x}) dx &= [x - e^{-x}]_0^1 \\ &= (1 - e^{-1}) - (0 - e^0) \\ &= 1 - e^{-1} + 1 \\ &= 2 - e^{-1} \\ &= 2 - \frac{1}{e}.\end{aligned}$$

Q 4

1991

Evaluate $\int_0^1 e^{4x} dx$.

$$\begin{aligned}\int_0^1 e^{4x} dx &= \left[\frac{1}{4} e^{4x} \right]_0^1 \\ &= \frac{1}{4} [e^{4x}]_0^1 \\ &= \frac{1}{4} (e^4 - e^0) \\ &= \frac{1}{4} (e^4 - 1).\end{aligned}$$

Q 5

1994

Given that $\frac{d}{dx}(e^{x^2}) = 2xe^{x^2}$, evaluate $\int_0^1 xe^{x^2} dx$.

$$\begin{aligned}\frac{d}{dx}(e^{x^2}) &= 2xe^{x^2} \\ \therefore \frac{1}{2} \cdot \frac{d}{dx}(e^{x^2}) &= xe^{x^2} \\ \therefore \int_0^1 xe^{x^2} dx &= \frac{1}{2} [e^{x^2}]_0^1 \\ &= \frac{1}{2}(e^1 - e^0) \\ &= \frac{1}{2}(e - 1).\end{aligned}$$

Q 6

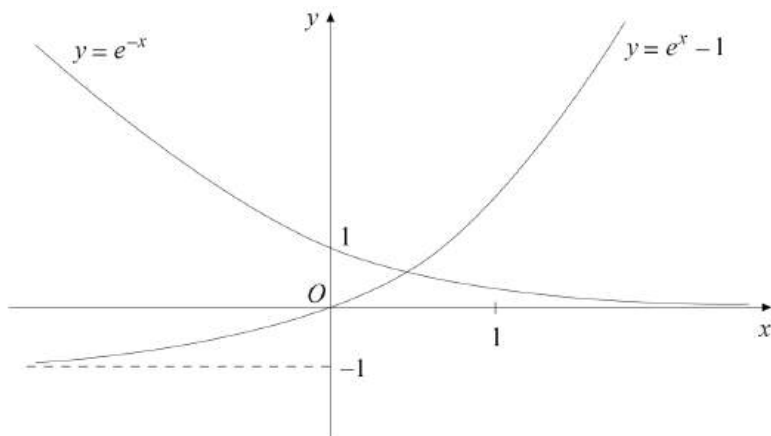
1997

Q 7

1996

Evaluate $\int_0^{\ln 7} e^{-x} dx$.

$$\begin{aligned} \int_0^{\ln 7} e^{-x} dx &= [-e^{-x}]_0^{\ln 7} \\ &= -e^{-\ln 7} - (-e^0) \\ &= -\frac{1}{e^{\ln 7}} - (-1) \\ &= -\frac{1}{7} + 1 \\ &= \frac{6}{7} \end{aligned}$$



The diagram shows the graphs of $y = e^x - 1$ and $y = e^{-x}$.

- Find the area between the curves from $x=1$ to $x=2$. Leave your answer in terms of e .
- Show that the curves intersect when $e^{2x} - e^x - 1 = 0$.
- Use the results of part (b) with $u = e^x$ to show that the x coordinate of the point of intersection of the curves is approximately 0.481.

$$\begin{aligned} \text{(i) } A &= \int_1^2 (e^x - 1) dx - \int_1^2 e^{-x} dx \\ &= \int_1^2 (e^x - 1 - e^{-x}) dx \\ &= [e^x - x + e^{-x}]_1^2 \\ &= (e^2 - 2 + e^{-2}) - (e^1 - 1 + e^{-1}) \\ &= e^2 - 2 + \frac{1}{e^2} - e + 1 - \frac{1}{e} \\ &= (e^2 + \frac{1}{e^2} - e - \frac{1}{e} - 1) \text{ units}^2 \end{aligned}$$

(ii) The curves intersect when

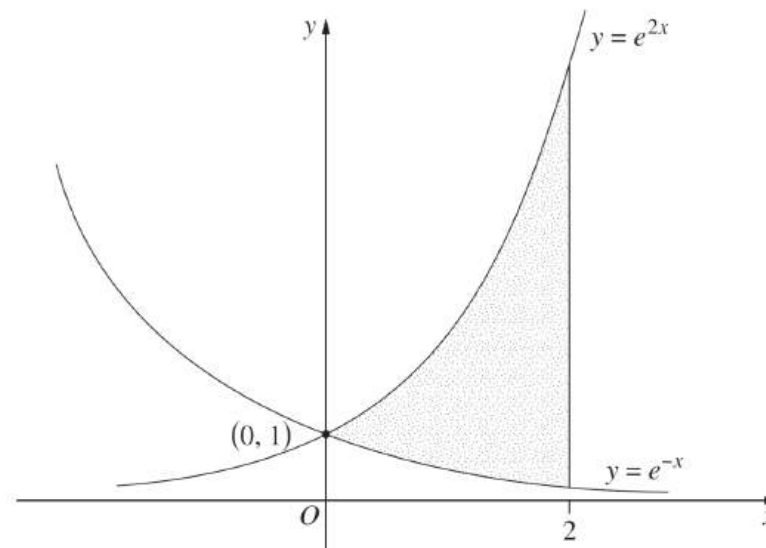
$$\begin{aligned} e^x - 1 &= e^{-x} \\ \therefore e^x - 1 &= \frac{1}{e^x} \\ \therefore e^{2x} - e^x &= 1 \\ \therefore e^{2x} - e^x - 1 &= 0. \end{aligned}$$

- From part (b), curves intersect at 1.618, but $u = e^x$
 $\therefore e^x = 1.618$
 $x = \log_e 1.618$
 $\therefore x \approx 0.481$ (by calc.)

Q 8

2010

The curves $y = e^{2x}$ and $y = e^{-x}$ intersect at the point $(0, 1)$ as shown in the diagram.



Find the exact area enclosed by the curves and the line $x = 2$.

Q4b Area of the enclosed region $= \int_0^2 (e^{2x} - e^{-x}) dx$

$$= \left[\frac{e^{2x}}{2} - \frac{e^{-x}}{-1} \right]_0^2 = \left(\frac{e^4}{2} + e^{-2} \right) - \left(\frac{1}{2} + 1 \right) = \frac{e^4}{2} + e^{-2} - \frac{3}{2}$$

Evaluate $\int_1^2 e^{3x} dx$

$$\begin{aligned} \int_1^2 e^{3x} dx &= \left[\frac{1}{3} e^{3x} \right]_1^2 \\ &= \frac{1}{3} e^6 - \frac{1}{3} e^3 \\ &= \frac{1}{3} (e^6 - e^3). \end{aligned}$$

Q 9

2004



extending calculus BOs

Part D: Integrating expo

Unit

NAME: _____

Q10

2014

Which expression is equal to $\int e^{2x} dx$?

(A) $e^{2x} + c$

(B) $2e^{2x} + c$

(C) $\frac{e^{2x}}{2} + c$

(D) $\frac{e^{2x+1}}{2x+1} + c$

$$\int e^{2x} dx = \frac{1}{2} e^{2x} + c$$

Q11

2015

Using the trapezoidal rule with 4 subintervals, which expression gives the approximate area under the curve $y = xe^x$ between $x = 1$ and $x = 3$?

(A) $\frac{1}{4}(e^1 + 6e^{1.5} + 4e^2 + 10e^{2.5} + 3e^3)$

(B) $\frac{1}{4}(e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$

(C) $\frac{1}{2}(e^1 + 6e^{1.5} + 4e^2 + 10e^{2.5} + 3e^3)$

(D) $\frac{1}{2}(e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$

$\frac{x}{y}$	1	1.5	2	2.5	3
	e	$1.5e^{1.5}$	$2e^2$	$2.5e^{2.5}$	$3e^3$

By trapezoidal rule,

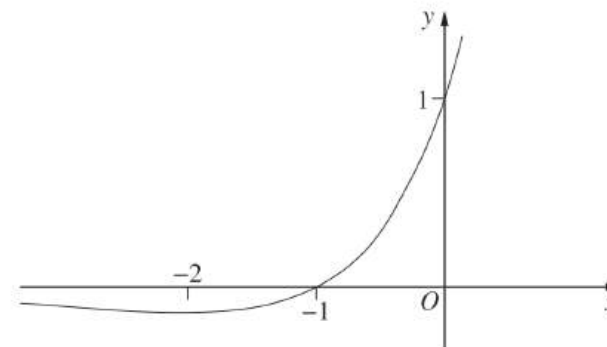
$$A \approx \frac{0.5}{2} (e + 2(1.5e^{1.5} + 2e^2 + 2.5e^{2.5}) + 3e^3) = \frac{1}{4} (e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$$

Answer: B

Q12

2015

The diagram shows the graph of $y = e^x(1+x)$.



How many solutions are there to the equation $e^x(1+x) = 1 - x^2$?

(A) 0

(B) 1

(C) 2

(D) 3

Draw the parabola (with intercepts $(0,1)$, $(-1,0)$, $(1,0)$) over the graph. There are two intersections.

Answer: C

Q13

2016

(i) Differentiate $y = xe^{3x}$.

(ii) Hence find the exact value of $\int_0^2 e^{3x}(3+9x)dx$.

Q12di $\frac{dy}{dx} = e^{3x} + 3xe^{3x} = e^{3x}(1+3x)$

Q12dii

$$\int_0^2 e^{3x}(3+9x)dx = 3 \times \int_0^2 e^{3x}(1+3x)dx = 3 \times [xe^{3x}]_0^2 = 3 \times 2e^6 = 6e^6$$



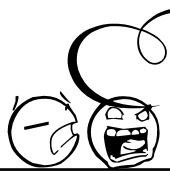
Q14

2018

Evaluate $\int_0^3 e^{5x} dx$.

$$\begin{aligned}\int_0^3 e^{5x} dx &= \left[\frac{1}{5} e^{5x} \right]_0^3 \\ &= \frac{1}{5} (e^{15} - e^0) \\ &= \frac{1}{5} (e^{15} - 1)\end{aligned}$$





extending calculus BOB

Part E: Integrating LOG

2 unit

NAME: _____

Q 1

1987

Find the exact value of: $\int_1^2 \frac{1}{1+x} dx$.

$$\begin{aligned} \int_1^2 \frac{1}{1+x} dx &= [\log_e(1+x)]_1^2 \\ &= \log_e 3 - \log_e 2 \\ &= \log_e \left(\frac{3}{2}\right) \\ &= \log_e 1.5. \end{aligned}$$

Q 2

1989

Find $\int (x^2 + \frac{2}{x}) dx$.

$$\int \left(x^2 + \frac{2}{x}\right) dx = \frac{x^3}{3} + 2 \ln x + C.$$

Q 3

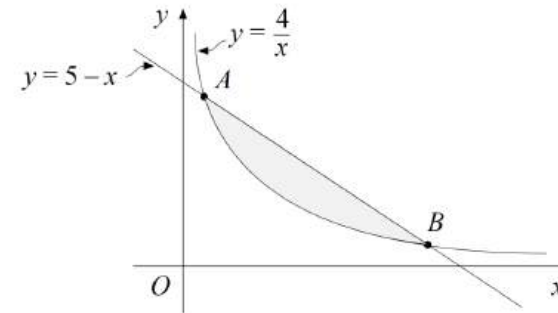
1991

Find $\int \frac{dx}{3x+5}$

$$\begin{aligned} \int \frac{dx}{3x+5} &= \frac{1}{3} \int \frac{3}{3x+5} dx \\ &= \frac{1}{3} \ln(3x+5) + C. \end{aligned}$$

Q 4

1995



NOT TO
SCALE

The diagram shows the graphs of $y = \frac{4}{x}$ and $y = 5 - x$.

The graphs intersect at the points A and B as shown.

- Find the x coordinates of the points A and B.
- Find the area of the shaded region between $y = \frac{4}{x}$ and $y = 5 - x$.

(i) Solving simultaneously:

$$\begin{aligned} \frac{4}{x} &= 5 - x \\ 4 &= 5x - x^2 \\ x^2 - 5x + 4 &= 0 \\ (x-4)(x-1) &= 0 \\ \therefore x &= 4 \text{ or } 1 \end{aligned}$$

(ii) Required Area

$$\begin{aligned} &= \int_1^4 (5-x) dx - \int_1^4 \frac{4}{x} dx \\ &= \int_1^4 \left(5-x-\frac{4}{x}\right) dx \\ &= \left[5x - \frac{x^2}{2} - 4 \ln x\right]_1^4 \\ &= (20 - 8 - 4 \ln 4) - \left(5 - \frac{1}{2} - 4 \ln 1\right) \\ &= (12 - 4 \ln 4) - \left(4\frac{1}{2}\right) \\ &= \left(7\frac{1}{2} - 4 \ln 4\right) \text{ units}^2. \end{aligned}$$



extending calculus BOB

Part e: Integrating LOG



Q 5

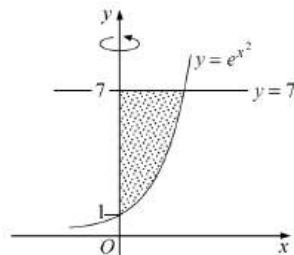
1998

Find $\int \frac{x}{x^2+3} dx$

$$\begin{aligned}\int \frac{x}{x^2+3} dx &= \frac{1}{2} \int \frac{2x}{x^2+3} dx \\ &= \frac{1}{2} \ln(x^2+3) + c.\end{aligned}$$

Q 6

1998



The shaded region bounded by $y = e^{x^2}$, $y = 7$ and the y axis is rotated around the y axis to form a solid.

- (i) Show that the volume of the solid is given by $V = \pi \int_1^7 \log_e y dy$.
- (ii) Copy and complete the table. Give your answers correct to 3 decimal places.

y	1	4	7
$\log_e y$			

- (iii) Use the trapezoidal rule with 3 function values to approximate the volume, V .

(i) $y = e^{x^2}$
 $x^2 = \log_e y$

Rotating about the y axis,

$$\begin{aligned}V &= \pi \int_1^7 x^2 dy \\ &= \pi \int_1^7 \log_e y dy.\end{aligned}$$

(ii)

y	1	4	7
$\log_e y$	0	1.386	1.946

(iii) $V = \pi \int_1^7 \log_e y dy$
 $\doteq \pi \times \frac{3}{2} [(0 + 1.386) + (1.386 + 1.946)]$
 $\therefore \text{Volume} \doteq 22.2 \text{ cubic units.}$

Q 7

2000

The area under the curve $y = \frac{1}{\sqrt{x}}$, for $1 \leq x \leq e^2$, is rotated about the x axis. Find the exact volume of the solid of revolution.

(a)
$$\begin{aligned}V &= \pi \int_1^{e^2} y^2 dx \\ &= \pi \int_1^{e^2} \left(\frac{1}{\sqrt{x}}\right)^2 dx \\ &= \pi \int_1^{e^2} \frac{1}{x} dx \\ &= \pi [\ln x]_1^{e^2} \\ &= \pi (\ln e^2 - \ln 1) \\ &= \pi (2 \ln e - \ln 1).\end{aligned}$$

 $\therefore \text{Volume} = 2\pi \text{ unit}^3.$

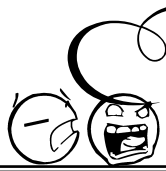
Q 8

2001

Evaluate $\int_0^1 \frac{dx}{x+4}$

$$\begin{aligned}\int_0^1 \frac{dx}{x+4} &= \ln(x+4) \Big|_0^1 \\ &= \ln 5 - \ln 4 \\ &= \ln \frac{5}{4}.\end{aligned}$$





extending calculus BOs

Part e: Integrating Log



Q 9

2004

Find $\int \frac{x}{x^2 - 3} dx$

$$\begin{aligned} \int \frac{x}{x^2 - 3} dx &= \frac{1}{2} \int \frac{2x}{x^2 - 3} dx \\ &= \frac{1}{2} \log_e(x^2 - 3) + C. \end{aligned}$$

Q10

2005

Find $\int \frac{6x^2}{x^3 + 1} dx$

$$\begin{aligned} \int \frac{f'(x)}{f(x)} dx &= \ln[f(x)] + C \\ \int \frac{6x^2}{x^3 + 1} dx &= 2 \int \frac{3x^2}{x^3 + 1} dx \\ &= 2 \ln(x^3 + 1) + C. \end{aligned}$$

Q11

2010

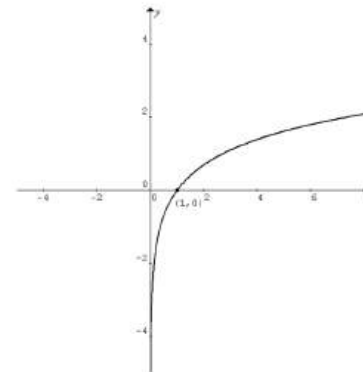
(i) Sketch the curve $y = \ln x$.

(ii) Use the trapezoidal rule with three function values to find an approximation to

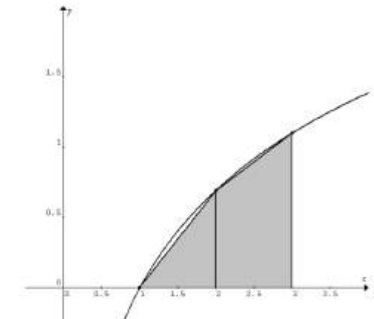
$$\int_1^3 \ln x dx.$$

(iii) State whether the approximation found in part (ii) is greater than or less than the exact value of $\int_1^3 \ln x dx$. Justify your answer.

Q3bi $y = \ln x$



Q3biii The approximation (total area of the two trapeziums) is less than the exact value of $\int_1^3 \ln x dx$ (the area under $y = \ln x$ between $x = 1$ and $x = 3$).



Q3bii $x = 1, y = 0; x = 2, y = \ln 2; x = 3, y = \ln 3$

$$\int_1^3 \ln x dx \approx \frac{1}{2} (0 + 2 \ln 2 + \ln 3)(1) = \ln 2 + \frac{1}{2} \ln 3$$



extending calculus BOs

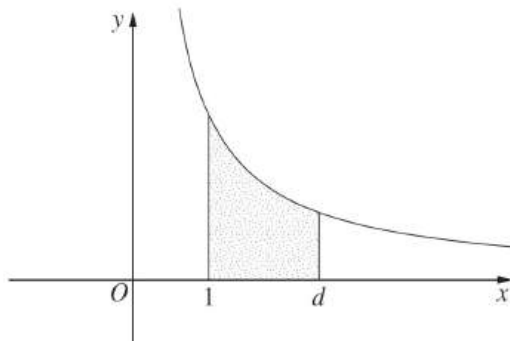
Part e: Integrating Log



Q12

2015

The diagram shows the area under the curve $y = \frac{2}{x}$ from $x = 1$ to $x = d$.



What value of d makes the shaded area equal to 2?

- (A) e
(B) $e + 1$
(C) $2e$
(D) e^2

$$\int_1^d \frac{2}{x} = 2 \Rightarrow 2[\ln x]_1^d = 2 \ln d = 2$$

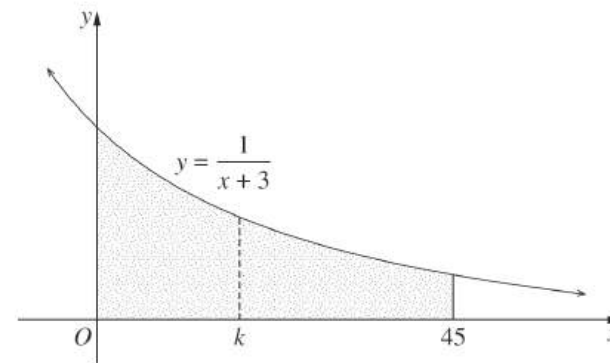
$$\therefore \ln d = 1 \Rightarrow d = e$$

Answer: A

Q13

2018

The diagram shows the region bounded by the curve $y = \frac{1}{x+3}$ and the lines $x = 0$, $x = 45$ and $y = 0$. The region is divided into two parts of equal area by the line $x = k$, where k is a positive integer.



NOT TO SCALE

What is the value of the integer k , given that the two parts have equal areas?

$$\int_0^k \frac{dx}{x+3} + \int_k^{45} \frac{dx}{x+3} = \int_0^{45} \frac{dx}{x+3}$$

$$\frac{(k+3)^2}{9} = 16$$

$$\int_0^k \frac{dx}{x+3} = \int_k^{45} \frac{dx}{x+3}$$

$$(k+3)^2 = 144$$

$$k+3 = \pm 12 \quad (k > 0)$$

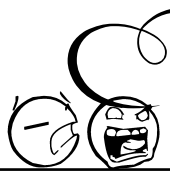
$$2 \int_0^k \frac{dx}{x+3} = \int_0^{45} \frac{dx}{x+3}$$

$$k+3 = 12$$

$$k = 9$$

$$2 \left[\ln(x+3) \right]_0^k = \left[\ln(x+3) \right]_0^{45}$$

$$2 \left[\ln(k+3) - \ln 3 \right] = \ln(48) - \ln 3$$



extending calculus BOS

Part E: Integrating Trig

2unit

NAME: _____

Q 1

2000

Find: $\int \sec^2 5x \, dx$

$$\int \sec^2 5x \, dx = \frac{1}{5} \tan 5x + C.$$

Q 2

2001

Find: $\int \cos 3x \, dx$

$$\int \cos 3x \, dx = \frac{1}{3} \sin 3x + C.$$

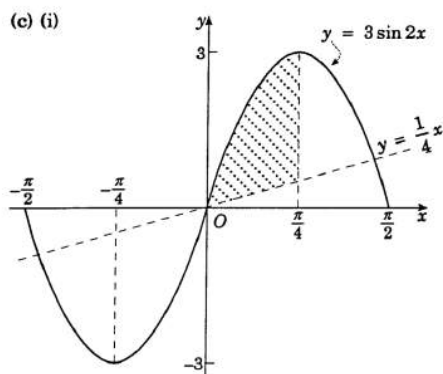
Q 3

2001

(i) Sketch the curve $y = 3 \sin 2x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.

(ii) On your diagram for part (i), sketch the line $y = \frac{1}{4}x$, and shade the region represented by $\int_0^{\frac{\pi}{4}} \left(3 \sin 2x - \frac{1}{4}x \right) dx$.

(iii) Find the exact value of the integral in part (ii).



(ii) See shaded region on the diagram.

$$\begin{aligned} \text{(iii)} \quad & \int_0^{\frac{\pi}{4}} 3 \sin 2x - \frac{1}{4}x \, dx \\ &= \left[-\frac{3}{2} \cos 2x - \frac{1}{8}x^2 \right]_0^{\frac{\pi}{4}} \\ &= \left[-\frac{3}{2} \cos \left(2 \times \frac{\pi}{4} \right) - \frac{1}{8} \times \left(\frac{\pi}{4} \right)^2 \right] \\ &\quad - \left[-\frac{3}{2} \cos(2 \times 0) - \frac{1}{8} \times 0^2 \right] \\ &= -\frac{3}{2} \cos \frac{\pi}{2} - \frac{\pi^2}{128} + \frac{3}{2} \cos 0 \\ &= -\frac{\pi^2}{128} + \frac{3}{2}. \end{aligned}$$

Q 4

2003

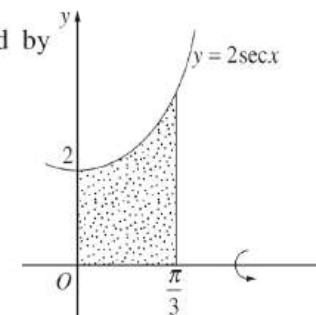
Evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sec^2 x \, dx$.

$$\begin{aligned} \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sec^2 x \, dx &= \left[\tan x \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}} \\ &= \tan \frac{\pi}{3} - \tan \frac{\pi}{4} \\ &= \sqrt{3} - 1. \end{aligned}$$

Q 5

2004

In the diagram, the shaded region is bounded by the curve $y = 2 \sec x$, the coordinate axes and the line $x = \frac{\pi}{3}$. The shaded region is rotated about the x -axis. Calculate the exact volume of the solid of revolution formed.



$$\begin{aligned} \text{Volume} &= \pi \int_a^b y^2 \, dx \\ &= \pi \int_0^{\frac{\pi}{3}} (2 \sec x)^2 \, dx \\ &= \pi \int_0^{\frac{\pi}{3}} 4 \sec^2 x \, dx \\ &= 4\pi \int_0^{\frac{\pi}{3}} \sec^2 x \, dx \\ &= 4\pi \left[\tan x \right]_0^{\frac{\pi}{3}} \\ &= 4\pi \left(\tan \frac{\pi}{3} - \tan 0 \right) \\ &= 4\pi (\sqrt{3} - 0) \\ &= 4\pi\sqrt{3} \text{ units}^3. \end{aligned}$$



extending calculus BOS

Part e: Integrating Trig

2 unit

NAME: _____

Q 6

2005

Find a primitive of $4 + \sec^2 x$. $\int (4 + \sec^2 x) dx = 4x + \tan x + C$.

Q 7

2005

Evaluate $\int_0^{\frac{\pi}{6}} \cos 3x dx$. $\int_0^{\frac{\pi}{6}} \cos 3x dx = \left[\frac{1}{3} \sin 3x \right]_0^{\frac{\pi}{6}}$
 $= \frac{1}{3} \left(\sin \frac{\pi}{2} - \sin 0 \right)$
 $= \frac{1}{3} (1 - 0)$
 $= \frac{1}{3}$.

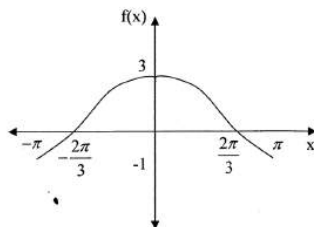
Q 8

2006

A function $f(x)$ is defined by $f(x) = 1 + 2 \cos x$.

- Show that the graph of $y = f(x)$ cuts the x -axis at $x = \frac{2\pi}{3}$.
- Sketch the graph of $y = f(x)$ for $-\pi \leq x \leq \pi$ showing where the graph cuts each of the axes.
- Find the area under the curve $y = f(x)$ between $x = -\frac{\pi}{2}$ and $x = \frac{2\pi}{3}$.

(b) (i)
 $f(x) = 1 + 2 \cos x$
 let $f(x) = 0$
 $0 = 1 + 2 \cos x$
 $\cos x = \frac{-1}{2}$
 $x = \frac{2\pi}{3}$



(ii)
 $f(-x) = 1 + 2 \cos(-x)$
 $= 1 + 2 \cos x = f(x) \therefore$ an even function

(iii)
 $A = \int_{-\frac{\pi}{2}}^{\frac{2\pi}{3}} 1 + 2 \cos x dx$
 $= \left[x + 2 \sin x \right]_{-\frac{\pi}{2}}^{\frac{2\pi}{3}}$
 $= \left[\frac{2\pi}{3} + 2 \sin \frac{2\pi}{3} - \left(-\frac{\pi}{2} + 2 \sin \left(-\frac{\pi}{2} \right) \right) \right]$
 $= \frac{2\pi}{3} + \sqrt{3} + \frac{\pi}{2} + 2$
 $= \frac{7\pi}{6} + \sqrt{3} + 2$

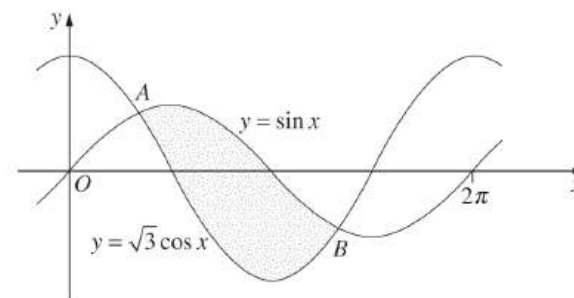
Q 9

2007

Find $\int (1 + \cos 3x) dx$. $\int (1 + \cos 3x) dx$
 $= x + \frac{1}{3} \sin 3x + c$

Q10

2007

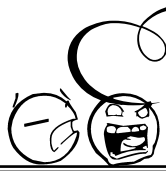


The diagram shows the graphs of $y = \sqrt{3} \cos x$ and $y = \sin x$. The first two points of intersection to the right of the y -axis are labelled A and B .

- Solve the equation $\sqrt{3} \cos x = \sin x$ to find the x -coordinates of A and B .
- Find the area of the shaded region in the diagram.

(b) (i) $\sqrt{3} \cos x = \sin x$
 $\sqrt{3} = \tan x$
 $\therefore x = \frac{\pi}{3}, \frac{4\pi}{3}$

(ii) $\int_{\frac{\pi}{3}}^{\frac{4\pi}{3}} \sin x - \sqrt{3} \cos x dx$
 $= \left[-\cos x - \sqrt{3} \sin x \right]_{\frac{\pi}{3}}^{\frac{4\pi}{3}}$
 $= \left(-\cos \frac{4\pi}{3} - \sqrt{3} \sin \frac{4\pi}{3} \right) - \left(-\cos \frac{\pi}{3} - \sqrt{3} \sin \frac{\pi}{3} \right)$
 $= \left(\frac{1}{2} + \sqrt{3} \frac{\sqrt{3}}{2} \right) - \left(-\frac{1}{2} - \sqrt{3} \frac{\sqrt{3}}{2} \right)$
 $= 2 + 2 = 4$



extending calculus BOB

Part E: Integrating Trig

Unit

NAME: _____

Q11

2008

The gradient of a curve is given by $\frac{dy}{dx} = 1 - 6\sin 3x$. The curve passes through the point $(0, 7)$.

What is the equation of the curve?

Q5a $\frac{dy}{dx} = 1 - 6\sin 3x$, $y = \int (1 - 6\sin 3x) dx = x + 2\cos 3x + c$.

The curve passes through $(0, 7)$, $\therefore 7 = 2\cos 0 + c$, $c = 5$.

Hence $y = x + 2\cos 3x + 5$.

Q12

2008

Evaluate $\int_0^{\frac{\pi}{12}} \sec^2 3x \, dx$.

$$\int_0^{\frac{\pi}{12}} \sec^2 3x \, dx = \left[\frac{1}{3} \tan 3x \right]_0^{\frac{\pi}{12}} = \frac{1}{3} \tan \frac{\pi}{4} = \frac{1}{3}$$

Q13

2008

(i) Differentiate $\log_e (\cos x)$ with respect to x .

(ii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \tan x \, dx$.

Q3bi $\frac{d}{dx} \log_e (\cos x) = \frac{1}{\cos x} \times (-\sin x) = -\tan x$

Q3bii $\int_0^{\frac{\pi}{4}} \tan x \, dx = [-\log_e (\cos x)]_0^{\frac{\pi}{4}} = -\log_e \left(\cos \frac{\pi}{4} \right) + \log_e (\cos 0)$
 $= -\log_e \left(\frac{1}{\sqrt{2}} \right) + 0 = \log_e (\sqrt{2}) = \frac{1}{2} \log_e 2$

Q14

2012

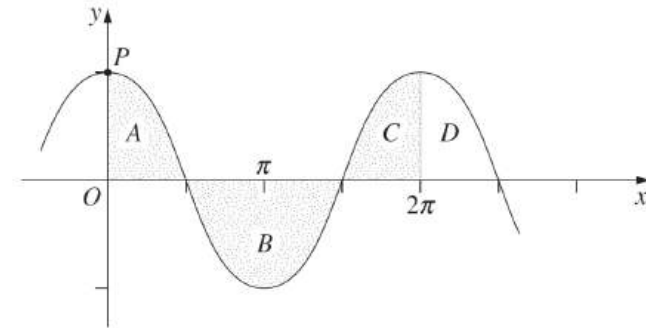
Find $\int_0^{\frac{\pi}{2}} \sec^2 \frac{x}{2} \, dx$.

$$\int_0^{\frac{\pi}{2}} \sec^2 \frac{x}{2} \, dx = 2 \left[\tan \frac{x}{2} \right]_0^{\frac{\pi}{2}} = 2 \tan \frac{\pi}{4} = 2$$

Q15

2011

The diagram shows the graph $y = 2 \cos x$.



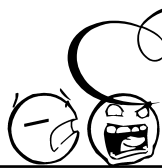
(i) State the coordinates of P .

(ii) Evaluate the integral $\int_0^{\frac{\pi}{2}} 2 \cos x \, dx$.

(iii) Indicate which area in the diagram, A , B , C or D , is represented by the integral $\int_{\frac{3\pi}{2}}^{2\pi} 2 \cos x \, dx$.

(iv) Using parts (ii) and (iii), or otherwise, find the area of the region bounded by the curve $y = 2 \cos x$ and the x -axis, between $x = 0$ and $x = 2\pi$.

(v) Using the parts above, write down the value of $\int_{\frac{\pi}{2}}^{2\pi} 2 \cos x \, dx$.



extending calculus BOs

Part e: Integrating Trig



NAME: _____

Q6ci $P(0,2)$

Q6cii $\int_0^{\frac{\pi}{2}} 2 \cos x dx = [2 \sin x]_0^{\frac{\pi}{2}} = 2 \sin \frac{\pi}{2} - 2 \sin 0 = 2$

Q6ciii C

Q6civ $Area = 2 \times 4 = 8$ unit squares

Q6cv $\int_{\frac{\pi}{2}}^{2\pi} 2 \cos x dx = -2 - 2 + 2 = -2$

Q16

2014

Evaluate $\int_0^{\frac{\pi}{2}} \sin \frac{x}{2} dx$.

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \sin \frac{x}{2} dx &= -2 \left[\cos \frac{x}{2} \right]_0^{\frac{\pi}{2}} \\ &= 2 \left(1 - \frac{1}{\sqrt{2}} \right) \\ &= 2 - \sqrt{2} \end{aligned}$$

Q17

2018

A trigonometric function $f(x)$ satisfies the condition

$$\int_0^{\pi} f(x) dx \neq \int_{\pi}^{2\pi} f(x) dx.$$

Which function could be $f(x)$?

- A. $f(x) = \sin(2x)$

B. $f(x) = \cos(2x)$

C. $f(x) = \sin\left(\frac{x}{2}\right)$

D. $f(x) = \cos\left(\frac{x}{2}\right)$